

Optimal Regularity for Alt-Phillips

Free Boundary Problem

Mark Ma

Columbia

What is Alt - Phillips Free Boundary Problem?

- We want to minimize for $\sigma \in (-2, 2)$.

$$J^\sigma(u) := \int_{B_1} |\nabla u|^2 + u^\sigma \chi_{\{u > 0\}}$$

among $u \in H^1(B_1)$. $u = 0 \geq 0$ on ∂B_1 .

- Minimization Problem corresponds to

"solving"
$$\begin{cases} \Delta u = \frac{\sigma}{2} u^{\sigma-1} = \frac{\sigma}{2} u^{\sigma-2} \cdot u & \{u > 0\} \\ \frac{|\nabla u|^2}{u^\sigma} = C_\sigma & \partial \{u > 0\} \end{cases}$$

for $\sigma \in (-2, 2)$, the free boundary $\partial \{u > 0\}$ exists.

- End Point cases.

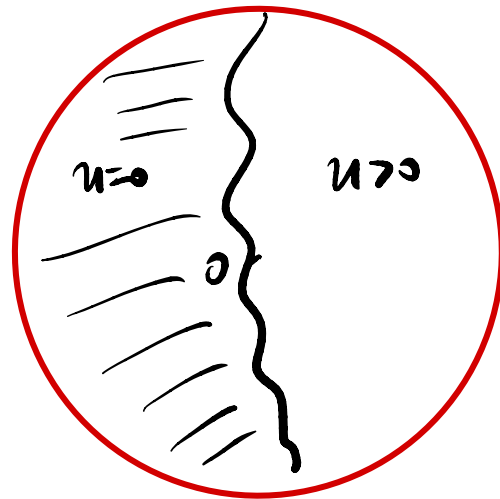
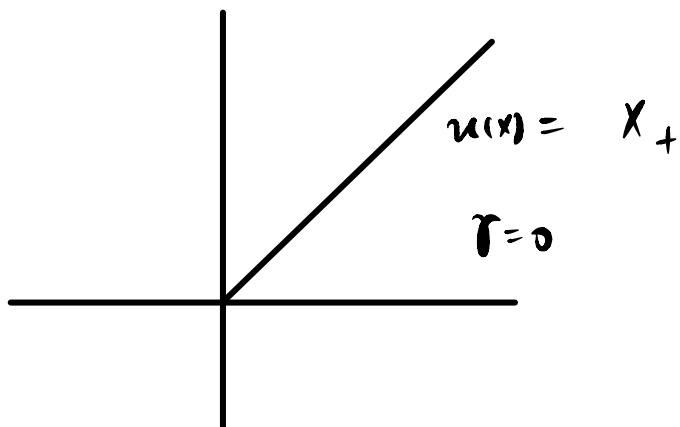
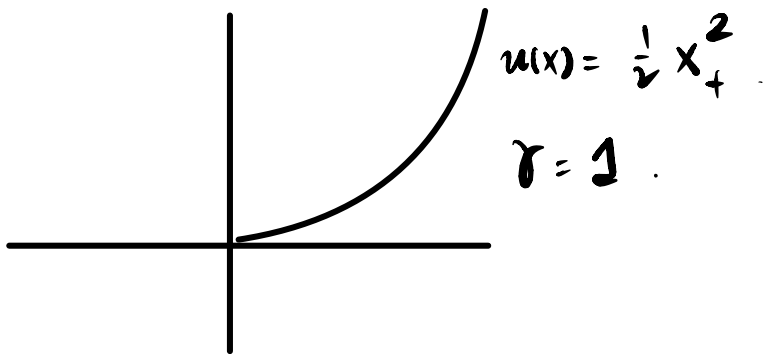
$\gamma = 1$ Obstacle Problem. (1977)

$$\Delta u = \chi_{\{u > 0\}}$$

$\gamma = 0$ One Phase Bernoulli (1981)

$$\begin{cases} \Delta u = 0 & \{u > 0\} \\ |\nabla u| = 1 & \partial \{u = 0\} \end{cases}$$

- One-dimensional profiles



What is the optimal Regularity for minimizers?

To see this, we conduct **Rescaling**

Let u minimize J^γ in B_r , we want to find $u_p(x) = \frac{1}{r^\beta} u(rx)$
that minimizes J^γ in B_1 .

Compute $J^\gamma(r^\beta u_p(\frac{x}{r}), B_1) = \int_{B_1} r^{2\beta-2} |\nabla u_p|^2 + r^{\gamma\beta} u_p^\gamma +$

equate $2\beta-2 = \gamma\beta \Rightarrow \beta = \frac{2}{2-\gamma}$

THM (1983 Phillips, 1986 Alt-Phillips)

Let $u \geq 0$ minimize $J^\gamma(u) = \int_{B_1} |\nabla u|^2 + u^\gamma$ for $\gamma \in (0,1)$

then $\|u\|_{C^{1,\beta}(\overline{B_{1/2}})} \leq C(n,\gamma) (\|u\|_{L^\infty(B_1)} + 1)$.

How to hit optimal regularity? Optimal Growth from Free Boundary

• $\gamma = 1 \quad \Delta u = \chi_{\{u > 0\}}$

Harnack $\sup_{B_\rho} u \leq C \cdot (\inf_{B_\rho} u + \rho^2) \leq C \rho^2 \text{ for } 0 \in \partial\{u > 0\}$

$$\| \partial^2 u \|_{L^\infty(B_\rho(x))} \leq C \frac{1}{\rho^2} (\| u \|_{L^\infty(B_\rho(x))} + \rho^2) \leq C$$

• $\gamma = 0 \quad \Delta u = 0 \quad \{u > 0\}$

Harnack if $\int_{\partial B_1} u \geq C \Rightarrow u > 0 \text{ in } B_1$

consider $x \in \{u > 0\}$. $\rho = \text{dist}(x, \{u = 0\})$

$$| \partial u(x) | \leq C \cdot \frac{1}{\rho} \int_{\partial B_\rho(x)} u \leq C \cdot \lim_{\varepsilon \rightarrow 0} \frac{1}{\rho + \varepsilon} \int_{\partial B_{\rho+\varepsilon}(x)} u = C$$

• for general $\gamma \in (0, 1]$ $\Delta u = \frac{\gamma}{2} u^{\gamma-1} \chi_{\{u>0\}}$.

Harnack if $\frac{1}{\int_{\partial B_r} u} \geq C \Rightarrow u \geq c_0 \int_{\partial B_r} u$ in $B_{r/2}$

consider $x \in \{u>0\}$, $2r = \text{dist}(x, \{u=0\})$

$$\| \nabla u \|_{C^{0, \beta-1}(B_{r/2}(x))} \leq C \cdot \frac{1}{\int_{\partial B_r} u} \left(\| u \|_{L^\infty(B_r(x))} + \int_{\partial B_r(x)} u^{\gamma-1} \right) \leq C$$

Since $\int_{\partial B_r} u^{\gamma-1} \cdot \int_{\partial B_r} u = 1$

So numbers
work out !

HOWEVER ! Harnack Inequality is Difficult to obtain !

Our approach using Dichotomy Argument let $\delta \in (0, 1)$

Theorem (2026 Ma) $\forall \zeta \in (0, \frac{1}{2}]$, $\forall \varepsilon > 0$,

$\exists \gamma(\varepsilon) \in (0, \frac{1}{2})$ and $a_0 = a_0(\varepsilon, \zeta) \geq 1$

s.t. $\forall$ u minimizer to $J^\delta(u, B_1)$

with $a = \left(\int_{B_1} u^2 \right)^{\frac{1}{2}} \geq a_0$.

Either (1) $\frac{1}{\zeta^p} \left(\int_{B_\zeta} u^2 \right)^{\frac{1}{2}} \leq \frac{1}{2} a$ L^2 average decrease by $\frac{1}{2}$

OR (2) $\exists m > 0$ s.t. $c_0(\zeta)a \leq m \leq C_1 a$

and $\left(\int_{B_\gamma} |u - m|^2 \right)^{\frac{1}{2}} \leq \varepsilon \cdot a$

u is sufficiently close to center of harmonic replacement.

If we're always in (1) case.

keep iterating yields $(\int_{B_{\frac{1}{2}}} u^2)^{\frac{1}{2}} \leq C \cdot \frac{1}{2}^{\beta} \quad \forall \frac{1}{2}$.

$\Rightarrow$ β -growth rate. and $0 \in \{u=0\}$.

But if at some step we're in (2) case.

we may iterate at α -scale, and choose $\varepsilon = \varepsilon(\alpha)$ small
to obtain $m_\alpha > 0$ s.t. $\frac{1}{2} C_0 \alpha < m_\alpha < 2 C_1 \alpha$

s.t. $(\int_{B_\rho} |u - m_\alpha|^2)^{\frac{1}{2}} \leq C_0 \cdot \alpha \cdot \rho^\alpha$.

$\Rightarrow C^{0,\alpha}$ Hölder continuity and $\frac{1}{4} C_0 \alpha \leq u(x) \leq 4 C_1 \alpha \quad \forall x \in B_{1/2}$

The similar method applies to
Multiple Membrane Att-Phillips Problem !
which is of our interest 😊.

THANK YOU !